

2. Ambiguity Map & Contradictions

AMB-001: Definition of 'Small and Marginal Farmer'

Schemes: PM-KISAN, KCC

Conflict: PM-KISAN defines eligibility around 'landholding farmer families' but does not set an explicit upper acreage limit after the Feb 2019 revision. KCC uses the RBI definition (up to 5 acres irrigated / 10 acres rain-fed). These may not align across states.

Severity: high

Resolution: Use state-level land ceiling data as the canonical threshold. Flag for human review when `land_owned_acres` is near the boundary.

AMB-002: Income thresholds overlap between PM-KISAN and PMAY-G

Schemes: PM-KISAN, PMAY-G

Conflict: PM-KISAN excludes income-tax payers (implying ~Rs 5-7 lakh threshold) while PMAY-G uses SECC deprivation criteria. A family could be eligible for PM-KISAN but ineligible for PMAY-G despite being economically weaker, because SECC criteria are multi-dimensional.

Severity: medium

Resolution: Evaluate each scheme independently. Do not infer PMAY-G eligibility from PM-KISAN status.

AMB-003: Age eligibility contradiction between PMEGP and DDU-GKY

Schemes: PMEGP, DDU-GKY

Conflict: PMEGP allows applicants aged 18+ with no upper limit for manufacturing. DDU-GKY targets youth aged 15-35. For a 34-year-old SC applicant interested in skill training AND enterprise setup, both schemes appear eligible but DDU-GKY is meant for 'youth' while PMEGP targets 'entrepreneurs' - the intended beneficiary persona may conflict.

Severity: low

Resolution: Allow dual eligibility. Present both schemes but note the different programme objectives to the user.

AMB-004: BPL classification inconsistency across 6 schemes

Kalam Welfare Engine - Official Documentation

Schemes: PMUY, NSAP, PMAY-G, DDU-GKY, NSAP-IGNWPS, NSAP-IGNDPS

Conflict: Multiple schemes reference 'BPL families' but the BPL list is outdated (2011 SECC). Some states use updated criteria while others still rely on the 2011 list. A user's `is_bpl` status may be valid in one state but not another. PMUY expanded eligibility to 7 additional categories beyond BPL.

Severity: high

Resolution: Accept self-declared BPL status but flag it for verification. Recommend the user check their name on the state BPL list.

AMB-005: Urban vs Rural classification for housing schemes

Schemes: PMAY-G, PMAY-U

Conflict: PMAY-G is for rural areas and PMAY-U is for urban. However, peri-urban areas and census towns that are administratively rural but demographically urban create eligibility gaps. A user in such an area might be rejected by both schemes.

Severity: high

Resolution: When `is_rural` and `is_urban` are both `None` or both `True`, flag the user's location for manual verification and show both schemes as potentially eligible.

AMB-006: Pension scheme overlap: NSAP vs APY for unorganized workers

Schemes: NSAP, APY

Conflict: APY targets unorganized workers aged 18-40 for future pension from age 60. NSAP provides immediate pension at age 60 for BPL. A worker who contributed to APY but also qualifies as BPL at 60 may be eligible for both - APY pension and NSAP-IGNOAPS pension. Whether both can be drawn simultaneously is unclear.

Severity: medium

Resolution: Flag dual-pension eligibility to the user. Some states prohibit drawing both APY and NSAP simultaneously.

AMB-007: Bank account requirement creates circular dependency with PMJDY

Schemes: PMJDY, PM-KISAN, PMUY, MGNREGA

Kalam Welfare Engine - Official Documentation

Conflict: PM-KISAN, PMUY, MGNREGA all require a bank account for DBT. PMJDY is the scheme to get a bank account. A user without a bank account would be 'ineligible' for all DBT schemes but 'eligible' for PMJDY. The engine correctly handles this, but it may confuse users.

Severity: medium

Resolution: When `has_bank_account` is False, proactively recommend PMJDY enrollment AND flag that DBT schemes become accessible after account opening.

AMB-008: Widow remarriage status and NSAP-IGNWPS eligibility

Schemes: NSAP-IGNWPS, PMMVY

Conflict: NSAP-IGNWPS provides pension to widows. If a beneficiary remarries, she is no longer a widow, but whether the pension stops immediately or continues is interpreted differently by states.

Similarly, PMMVY maternity benefit applies differently if the mother was previously widowed.

Severity: medium

Resolution: Use current `marital_status` and `is_widow` as the canonical check. Flag the remarriage edge case for human review.

AMB-009: Loan schemes overlap: PMMY vs PMEGP for micro-enterprises

Schemes: PMMY, PMEGP

Conflict: Both PMMY (MUDRA) and PMEGP target new micro-enterprises. PMMY offers loans up to Rs 10L. PMEGP offers subsidy-linked credit up to Rs 50L. A new entrepreneur could theoretically apply to both. PMEGP explicitly excludes existing PMMY beneficiaries from PMEGP subsidy.

Severity: medium

Resolution: Show both as eligible but note that PMEGP subsidy cannot be combined with an active MUDRA loan. Users should choose one path.

AMB-010: Occupation field semantic mismatch across schemes

Schemes: PM-KISAN, PMMY, PM-SVANidhi, MGNREGA

Conflict: PM-KISAN excludes 'institutional_landholder' (an occupation value), PMMY checks for non-farm activity, PM-SVANidhi requires 'street_vendor', and MGNREGA accepts any willing worker. The occupation field cannot simultaneously be 'farmer' and 'street_vendor'. Users with multiple

Kalam Welfare Engine - Official Documentation

livelihoods are forced into a single classification.

Severity: high

Resolution: Consider supporting an occupation list rather than single value. For now, match against primary occupation and flag users with mixed livelihoods.

AMB-011: Housing ownership: PMAY requires non-ownership while KCC implies land ownership

Schemes: PMAY-G, PMAY-U, KCC

Conflict: PMAY-G and PMAY-U require that the beneficiary does NOT own a pucca house. KCC requires land ownership for farming. A farmer with agricultural land but no house is eligible for both KCC and PMAY-G.

However, if they get a house through PMAY, their asset status changes for other schemes.

Severity: low

Resolution: Evaluate independently. owns_house and land_owned_acres are separate profile fields that can coexist.

AMB-012: Age boundaries create cliff effects across social security schemes

Schemes: DDU-GKY, APY, NSAP, NSAP-IGNWPS

Conflict: DDU-GKY caps at 35, APY enrollment closes at 40, NSAP pension starts at 60, IGNWPS starts at 40. A person aged 36-39 falls in a gap: too old for DDU-GKY but still eligible for APY. At 40, APY closes but IGNWPS opens (if widow). At 60, NSAP opens. Age boundaries create 'dead zones' where no scheme targets the user.

Severity: medium

Resolution: When a user falls in an age gap, explicitly inform them of upcoming eligibility (e.g., 'You will become eligible for NSAP in X years').